The Mole & Molar Mass

Amedeo Avagadro

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The mass of a single atom, or even molecule, is incredibly small. So small, that using the mass of single atom has little purpose in real-world applications.

In 1811, Amedeo Avagadro determined that any convenient quantity of matter must contain an enormous number of atoms, ions, molecules, etc.

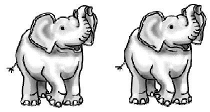
Using carbon as an example, he showed that 6.02×10^{23} atoms of carbon had a mass of 12.01 g. This value of 12.01 is significant, as it is the numerical value of the element's atomic mass.

This remains true for any element or molecule you choose. NaCl has an atomic mass of 58.44 (22.99 + 35.45). When you measure out 6.02×10^{23} molecules of NaCl, you find that it has a mass of 58.44 g.

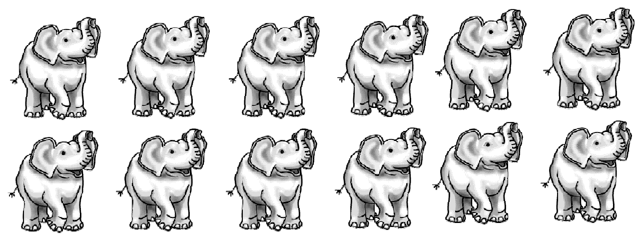
6.02×10^{23} is known as **Avagadro's Number (N_A)**. If we look at a quantity of items equal to 6.02×10^{23}, we call this a **mole**.

A mole is just quick way of summarizing a large quantities.

A pair (2) of elephants:



A dozen (12) elephants:



A mole (6.02×10^{23}) of elephants:

um.... never mind

Therefore the number of moles (n) present is equal to the ratio of the sample mass (m) to the molar mass (M)

$$n = \frac{m}{M}$$

C 12 g/mol
O 2 x 16 g/mol
44 g/mol

Ex) How many moles are present in 72.6 g of CO_2?

$n = ?$
$m = 72.6 g$
$M = 44 g/mol$

$$n = \frac{m}{M}$$

$$= \frac{72.6 \cancel{g}}{44 \cancel{g}/mol}$$

$$= 1.6 \text{ mol}$$

From the previous question, how many molecules of CO_2 are there?
How many carbon atoms? Oxygen atoms?

1.6 mol CO_2

$C = 1 \times 1.6\ mol = 1.6\ mol$

$O = 2 \times 1.6\ mol = 3.2\ mol$

# of molecules $= n \times N_A$

$C = 9.632 \times 10^{23}$

$= 9.6 \times 10^{23}$

$O = 1.9264 \times 10^{24}$

$= 1.9 \times 10^{24}$

The **Molar Mass (M)** of a substance is defined as the mass of one mole of that substance.

Carbon has a molar mass of 12.01 g/mol

Molecular Oxygen has a molar mass of

16 g/mol O $O_{2(g)}$ 32.0 g/mol

Water has a molar mass of

H_2O

H 2×1 g/mol = 2 = 18 g/mol

O 1×16 g/mol = +16

Ex) What is the mass of 1.7 mol of $C_6H_{12}O_6$?

$$C \quad 6 \times 12\ g/mol = 72\ g/mol$$
$$H \quad 12 \times 1.0\ g/mol = 12\ g/mol$$
$$O \quad 6 \times 16\ g/mol = 96\ g/mol$$
$$\overline{180\ g/mol}$$

$$= 180\ g/\cancel{mol}\,(1.7\ \cancel{mol})$$
$$= 306\ g$$
$$= 310\ g$$
$$= 3.1 \times 10^{2}\ g$$

Ex) Calculate the molar mass of lead, if 2.389 mol has a mass of 495 g.

$$n = \frac{m}{M}$$
$$M = \frac{m}{n}$$
$$M = \frac{495\ g}{2.389\ mol}$$
$$M = 207.199\ g/mol$$
$$= 207\ g/mol$$